

Assignment 1: Simple Linear Regression

TV Advertising Budget vs. Sales

Course: MD5001

1. Objective

This report performs a simple linear regression analysis to examine the relationship between TV advertising budget and Sales, using the dataset **tv_sales_data.csv**. The fitted model is:

$$\text{Sales} = \beta_0 + \beta_1 \cdot \text{TV} + \varepsilon$$

2. Dataset

The dataset contains 6 observations pairing TV advertising budget (in \$1,000s) with resulting Sales (in 1,000s of units):

TV (in \$1000s)	Sales (in 1000s of units)
5	7
10	9
15	12
20	14
25	18
30	21

3. Methodology and Code

The analysis was performed in Python using **pandas** for data handling and **statsmodels** (Ordinary Least Squares) to fit the regression model. Full code and executed output are provided in the accompanying notebook **assignment1_regression.ipynb**. Key steps:

```
import pandas as pd
import statsmodels.api as sm

df = pd.read_csv("tv_sales_data.csv")
X = sm.add_constant(df["TV"]) # adds intercept term (beta0)
y = df["Sales"]

model = sm.OLS(y, X).fit()
print(model.summary())
```

4. Results

Statistic	Intercept (β_0)	Slope – TV (β_1)
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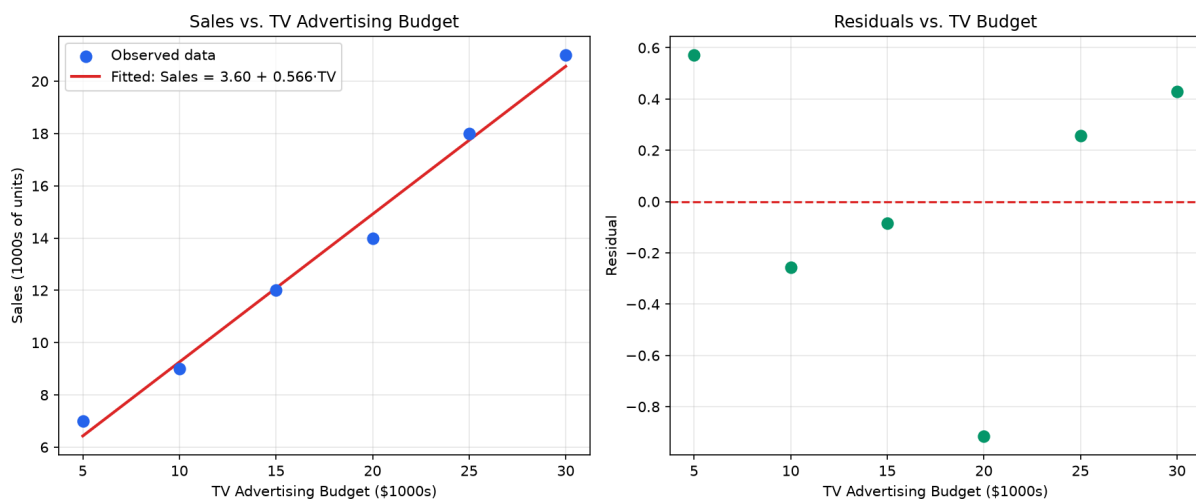
Estimated coefficient	3.6000	0.5657
Standard error	0.5674	0.0291
t-statistic	6.345	19.415
p-value	0.0032	4.15×10^{-5}

Fitted equation: Sales = 3.6000 + 0.5657 · TV

R-squared: 0.9895 **Adjusted R-squared:** 0.9869

F-statistic: 377.0 on 1 and 4 degrees of freedom **Prob (F-statistic):** 4.15×10^{-5}

5. Visualization



Left: observed data with fitted regression line. Right: residuals vs. TV budget, showing no strong systematic pattern.

6. Interpretation

Sign and magnitude of β_1

The slope β_1 (approximately 0.5657) is positive, indicating a positive relationship between TV advertising spend and sales: as the TV budget increases, sales increase. Specifically, each additional \$1,000 spent on TV advertising is associated with an estimated increase of about 566 units sold (0.5657 thousand units), on average, within the range of the observed data.

Statistical significance

The p-value for β_1 is approximately 4.15×10^{-5} , far below conventional significance thresholds such as 0.05 or 0.01. This provides strong statistically significant evidence against the null hypothesis $H_0: \beta_1 = 0$, supporting a real relationship between TV advertising and sales. The intercept β_0 is also statistically significant (p is approximately 0.0032), though it should be interpreted cautiously: it represents the model's predicted sales at a TV budget of \$0, which falls outside the observed data range (\$5,000–\$30,000) and is therefore more of a mathematical baseline than a directly meaningful real-world value.

Proportion of variance explained

The R-squared value of 0.9895 indicates that approximately 98.95% of the variability in sales is explained by the linear relationship with TV advertising budget in this sample, leaving only about 1.05% unexplained by the model. This is a very high R-squared, reflecting the strongly linear, low-noise nature of this particular (small, $n = 6$) dataset; real-world advertising data typically shows more scatter and a lower R-squared.

7. Conclusion

Based on this analysis, TV advertising budget is a strong, statistically significant, positive predictor of sales. The fitted model ($\text{Sales} = 3.60 + 0.5657 \cdot \text{TV}$) explains nearly all of the variation in sales observed in this dataset ($R^2 = 0.9895$), and both the slope and intercept are statistically significant at the 5% level. These results support the conclusion that, within this dataset, increased TV advertising spending is associated with meaningfully higher sales.